

References

- [1] Nanni, A., De Luca, A., and Zadeh, H. J.: Reinforced Concrete with FRP Bars: Mechanics and Design. CRC Press (2014). <https://doi.org/10.1201/b16669>
- [2] Bank, L. C.: Composites for construction: Structural design with FRP materials. John Wiley and Sons (2006). <https://doi.org/10.1002/9780470121429>
- [3] Emara, M., Salem, M. A., Mohamed, H. A., Shehab, H. A., and El Zohairy, A.: Shear Strengthening of Reinforced Concrete Beams Using Engineered Cementitious Composites and Carbon Fiber Reinforced Polymer Sheets. *Fibers* 11(11), 98 (2023). <https://doi.org/10.3390/fib11110098>
- [4] Peng, F., Xue, W., and Xue, W.: Database Evaluation of Shear Strength of Slender Fiber Reinforced Polymer Reinforced Concrete Members. *ACI Struct. J.* 117(3), 273–281 (2020). <https://doi.org/10.14359/51723504>
- [5] Zhang, Z., and Lan, J.: Study on Shear Behavior of Reinforced Concrete Beams Strengthened with FRP Grid–PCM Composite Reinforcement. *Appl. Sci.* 15(11), 6103 (2025). <https://doi.org/10.3390/app15116103>
- [6] Lu, D., Wang, X., Zhou, J., Chen, Z., Liu, P., Huang, H., and Wu, Z.: Hybrid XGBoost framework for predicting shear capacity of FRP grid strengthened RC beams. *Mater. Today Commun.* 49, 113980 (2025). <https://doi.org/10.1016/j.mtcomm.2025.113980>
- [7] El-Gamal, S., El-Salakawy, E., and Benmokrane, B.: Behavior of concrete bridge deck slabs reinforced with FRP bars under concentrated loads. *ACI Struct. J.* 102(5), 723–735 (2005).
- [8] Guadagnini, M., Pilakoutas, K., and Waldron, P.: Shear resistance of FRP RC beams. *J. Compos. Constr.* 10(6), 464–473 (2006). [https://doi.org/10.1061/\(ASCE\)1090-0268\(2006\)10:6\(464\)](https://doi.org/10.1061/(ASCE)1090-0268(2006)10:6(464))

[9] ACI 440.1R-15 – Guide for the Design and Construction of Structural Concrete Reinforced with FRP Bars

[10] Muttoni, A., and Ruiz, M. F.: Shear strength of members without transverse reinforcement as a function of critical shear crack width. *ACI Struct. J.* 105(2), 163–172 (2008). <https://doi.org/10.14359/19731>

[11] Naser, M. Z., Hawileh, R. A., and Abdalla, J. A.: Fiber reinforced polymer composites in strengthening reinforced concrete structures: A critical review. *Eng. Struct.* 198, 109542 (2019). <https://doi.org/10.1016/j.engstruct.2019.109542>

[12] Wang, C., Zou, X., Sneed, L. H., Zhang, F., Zheng, K., Xu, H., and Li, G.: Shear strength prediction of FRP strengthened concrete beams using interpretable machine learning. *Constr. Build. Mater.* 407(3), 133553 (2023). <https://doi.org/10.1016/j.conbuildmat.2023.133553>

[13] Ma, S., Sun, J., and Xu, T.: Numerical analysis of shear contribution of CFRP strengthened RC beams by different bond slip models. *Int. J. Concr. Struct. Mater.* 19(4) (2025). <https://doi.org/10.1186/s40069-024-00728-2>

[14] Al Saawani, M. A., Al Negheimish, A. I., El Sayed, A. K., and Alhozaimy, A. M.: Finite element modeling of debonding failures in FRP strengthened concrete beams using cohesive zone model. *Polymers* 14(9), 1889 (2022). <https://doi.org/10.3390/polym14091889>

[15] Huang, Q., Wang, W.-W., Tang, Y., Guo, K., and Zhou, C.: Failure modes analysis of reinforced concrete beams under impact loads based on machine learning and SHAP approach. *Eng. Fail. Anal.* 183, 110281 (2026). <https://doi.org/10.1016/j.engfailanal.2025.110281>

[16] Zhao, J., Zhu, M., Xu, L., Chen, M., and Shi, M.: Prediction of shear capacity of fiber reinforced polymer reinforced concrete beams based on machine learning. *Buildings* 15(11), 1908 (2025). <https://doi.org/10.3390/buildings15111908>

- [17] Yang, Y., and Liu, G.: Data-driven shear strength prediction of FRP reinforced concrete beams without stirrups based on machine learning methods. *Buildings* 13(2), 313 (2023). <https://doi.org/10.3390/buildings13020313>
- [18] Di, B., Qin, R., Zheng, Y., and Lv, J.: Investigation of the shear behavior of concrete beams reinforced with FRP rebars and stirrups using ANN hybridized with genetic algorithm. *Polymers* 15(13), 2857 (2023). <https://doi.org/10.3390/polym15132857>
- [19] Dinh, N. H., Kim, S.-H., and Choi, K.-K.: Evaluation of KDS 14 draft model for one-way shear strength of slender concrete members reinforced with fiber reinforced polymer (FRP) rebars. *Mater.* 19(63) (2025). <https://doi.org/10.1186/s40069-025-00804-1>
- [20] Nakamura, H., and Higai, T.: Evaluation of Shear Strength of Concrete Beams Reinforced with FRP. *Doboku Gakkai Ronbunshu* 26(508), 89–100 (1995). https://doi.org/10.2208/jscej.1995.508_89
- [21] Maruyama, K., and Zhao, W. J.: Size effect in shear behavior of FRP reinforced concrete beams. In: *Proc. 2nd Int. Conf. Adv. Compos. Mater. Bridges Struct.*, Canadian Soc. Civ. Eng., Montreal, QC, Canada, 227–234 (1996).
- [22] Vijay, P. V., and GangaRao, H. V. S.: A unified limit state approach using deformability factors in concrete beams reinforced with GFRP bars. In: *Proc. Materials for the New Millennium Conf.*, 657–665 (1996).
- [23] Mizukawa, Y., Sato, Y., Ueda, T., and Kakuta, Y.: A study on shear fatigue behavior of concrete beams with FRP rods. In: *Proceedings of the 3rd International Symposium on Non-Metallic (FRP) Reinforcement for Concrete Structures (FRPRCS-3)*, Sapporo, Japan, 309–316 (1997). Japan Concrete Institute.

- [24] Michaluk, C. R., Rizkalla, S. H., Tadros, G., and Benmokrane, B.: Flexural behavior of one-way concrete slabs reinforced by fiber reinforced plastic reinforcements. *ACI Struct. J.* 95(3), 353–365 (1998). <https://doi.org/10.14359/552>
- [25] Deitz, D. H., Harik, I. E., and Gesund, H.: One-way slabs reinforced with glass fiber reinforced polymer reinforcing bars. In: *Proc. 4th Int. Symp. Fiber Reinforced Polymer Reinforcement for Reinforced Concrete Structures (FRPRCS-4)*, ACI SP-188, 279–286 (1999). <https://doi.org/10.14359/5629>
- [26] Alkhrdaji, T., Wideman, M., Belarbi, A., and Nanni, A.: Shear strength of GFRP RC beams and slabs. In: *Proc. Int. Conf. Composites in Construction*, Balkema, Rotterdam, The Netherlands, 409–414 (2001).
- [27] Yost, J. R., Gross, S. P., and Dinehart, D. W.: Shear strength of normal strength concrete beams reinforced with deformed GFRP bars. *J. Compos. Constr.* 5(4), 268–275 (2001). [https://doi.org/10.1061/\(ASCE\)1090-0268\(2001\)5:4\(268\)](https://doi.org/10.1061/(ASCE)1090-0268(2001)5:4(268))
- [28] Tureyen, A. K., and Frosch, R. J.: Shear tests of FRP reinforced concrete beams without stirrups. *ACI Struct. J.* 99(4), 427–434 (2002). <https://doi.org/10.14359/12111>
- [29] Gross, S. P., Yost, J. R., Dinehart, D. W., Svensen, E., and Liu, N.: Shear strength of normal and high-strength concrete beams reinforced with GFRP bars. In: *Proc. Int. Conf. High Perform. Mater. Bridges*, ASCE, 426–437 (2003). [https://doi.org/10.1061/40691\(2003\)38](https://doi.org/10.1061/40691(2003)38)
- [30] Tariq, M., and Newhook, J. P.: Shear testing of FRP reinforced concrete without transverse reinforcement. In: *Proc. Annual Conf. Canadian Society for Civil Engineering (CSCE)*, Moncton, New Brunswick, Canada, 1330–1339 (2003).

- [31] Gross, S. P., Dinehart, D. W., Yost, J. R., and Theisz, P. M.: Experimental tests of high-strength concrete beams reinforced with CFRP bars. In: Proc. 4th Int. Conf. Advanced Composite Materials in Bridges and Structures (ACMBS IV), Calgary, Alberta, Canada, 1–8 (2004).
- [32] Razaqpur, A. G., Isgor, B. O., Greenaway, S., and Selley, A.: Concrete contribution to the shear resistance of fiber reinforced polymer reinforced concrete members. *J. Compos. Constr.* 8(5), 452–460 (2004). [https://doi.org/10.1061/\(ASCE\)1090-0268\(2004\)8:5\(452\)](https://doi.org/10.1061/(ASCE)1090-0268(2004)8:5(452))
- [33] Kilpatrick, A. E., and Easden, L. R.: Shear capacity of GFRP reinforced high-strength concrete slabs. In: *Developments in Mech. Struct. Mater.*, Vol. 1, A. J. Deeks and H. Hao (eds.), Taylor and Francis, London, 119–124 (2005).
- [34] El Sayed, A. K., El-Salakawy, E. F., and Benmokrane, B.: Shear strength of FRP reinforced concrete beams without transverse reinforcement. *ACI Struct. J.* 103(2), 235–243 (2006). <https://doi.org/10.14359/15181>
- [35] El Sayed, A. K., El-Salakawy, E. F., and Benmokrane, B.: Shear capacity of high-strength concrete beams reinforced with FRP bars. *ACI Struct. J.* 103(3), 383–389 (2006). <https://doi.org/10.14359/15316>
- [36] Guadagnini, M., Pilakoutas, K., and Waldron, P.: Shear resistance of FRP reinforced concrete beams: experimental study. *J. Compos. Constr.* 10(6), 464–473 (2006). [https://doi.org/10.1061/\(ASCE\)1090-0268\(2006\)10:6\(464\)](https://doi.org/10.1061/(ASCE)1090-0268(2006)10:6(464))
- [37] Kilpatrick, A. E., and Dawborn, R.: Flexural shear capacity of high-strength concrete slabs reinforced with longitudinal GFRP bars. In: Proc. FIB Conf., Naples, Italy, 1–10 (2006).
- [38] Matta, F., Nanni, A., Hernandez, T. M., and Benmokrane, B.: Scaling of strength of FRP reinforced concrete beams without shear reinforcement. In: Proc. 4th Int. Conf. FRP Compos. Civ. Eng. (CICE 2008), Zurich, Switzerland, Paper R22 (2008).

- [39] Niewels, J.: Zum Tragverhalten von Betonbauteilen mit Faserverbundkunststoff-Bewehrung. Ph.D. dissertation, RWTH Aachen University, Aachen, Germany (2008).
- [40] Steiner, S., El-Sayed, A. K., Benmokrane, B., and Matta, F.: Shear strength of large-size concrete beams reinforced with glass fiber reinforced polymer (GFRP) bars. In: Proc. 5th Int. Conf. Adv. Compos. Mater. Bridges Struct., Winnipeg, Canada, Paper ID 129 (2008).
- [41] Alam, M. S., Hussein, A., and Ebrahim, E. A. A.: Shear strength of concrete beams reinforced with glass fiber reinforced polymer (GFRP) bars. In: Proc. 9th Int. Symp. FRP Reinforcement Concrete Struct. (FRPRCS-9), Sydney, Australia (2009).
- [42] Alam, M. S., and Hussein, A.: Shear strength of concrete beams reinforced with glass fibre reinforced polymer (GFRP) bars. In: Proc. 9th Int. Symp. FRP Reinforcement Concrete Struct. (FRPRCS-9), Adelaide, Australia (2009).
- [43] Wang, H. S., Kim, M. S., Cho, J. M., and Kim, C. H.: Concrete shear strength of beams reinforced with FRP bars according to flexural reinforcement ratio and shear span-to-depth ratio. In: Proc. 9th Int. Symp. FRPRCS-9, Sydney, Australia (2009).
- [44] Caporale, A., and Luciano, R.: Indagine sperimentale e numerica sulla resistenza a taglio di travi di calcestruzzo armate con barre di GFRP. In: Proc. XXXVIII Convegno Nazionale AIAS, Turin, Italy (2009).
- [45] Bentz, E. C., Massam, L., and Collins, M. P.: Shear strength of large concrete members with FRP reinforcement. *J. Compos. Constr.* 14(6), 637–646 (2010).
[https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000108](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000108)
- [46] Maruyama, K., and Zhao, W. J.: Flexural and shear behaviour of concrete beams reinforced with FRP rods. In: Swamy, R. N. (ed.), *Corrosion and Corrosion Protection of Steel in Concrete*, Sheffield Academic Press, 1330–1339 (1994).

- [47] Zhao, W., Maruyama, K., and Suzuki, H.: Shear behavior of concrete beams reinforced by FRP rods as longitudinal and shear reinforcement. In: Proc. 2nd Int. RILEM Symp. Non-Metallic (FRP) Reinforcement Concrete Struct., Ghent, Belgium, 352–359 (1995).
- [48] Alsayed, S. H., Al-Salloum, Y. A., and Almusallam, T. H.: Shear design for beams reinforced by GFRP bars. In: Proc. 3rd Int. Symp. Non-Metallic (FRP) Reinforcement Concrete Struct., Sapporo, Japan, 285–292 (1997).
- [49] Duranovic, N., Pilakoutas, K., and Waldron, P.: Tests on concrete beams reinforced with glass fibre reinforced plastic bars. In: Proc. FRPRCS-3, Sapporo, Japan, 479–486 (1997).
- [50] Shehata, E., Morphy, R., and Rizkalla, S.: Fibre reinforced polymer shear reinforcement for concrete members: behaviour and design guidelines. *Can. J. Civ. Eng.* 27(5), 859–872 (2000).
<https://doi.org/10.1139/l00-004>
- [51] Alkhrdaji, T., Wideman, M., Belarbi, A., and Nanni, A.: Shear strength of GFRP RC beams and slabs. In: Proc. Int. Conf. Composites in Construction (CCC), Porto, Portugal, 409–414 (2001).
- [52] Ascione, L., Razaqpur, A. G., and Spadea, S.: Effectiveness of FRP stirrups in concrete beams subject to shear. In: Proc. 7th Int. Conf. Fibre-Reinforced Polymer (FRP) Composites in Civil Engineering (CICE 2014), Vancouver, Canada, 368–373 (2014).
- [53] Spadea, S.: Structural behaviour of FRP reinforced concrete members. Ph.D. thesis, University of Salerno, Italy (2010).
- [54] Tottori, S., and Wakui, H.: Shear capacity of RC and PC beams using FRP reinforcement. In: ACI SP-138, American Concrete Institute, Farmington Hills, MI, USA, 615–632 (1993).
- [55] Massam, L.: The behaviour of GFRP reinforced concrete beams in shear. Master's thesis, University of Toronto, Canada (2001).

- [56] Ali, I., Thamrin, R., Abdul, A. A., and Noridah, M.: Diagonal shear cracks and size effect in concrete beams reinforced with GFRP bars. *Appl. Mech. Mater.* 621, 113–119 (2014). <https://doi.org/10.4028/www.scientific.net/AMM.621.113>
- [57] Mohamed Said, M. A., Adam, M. A., Mahmoud, A. A., and Shanour, A. S.: Experimental and analytical shear evaluation of concrete beams reinforced with GFRP bars. *Constr. Build. Mater.* 102, 574–591 (2016). <https://doi.org/10.1016/j.conbuildmat.2015.10.185>
- [58] Vora, T. P., and Shah, B. J.: Experimental investigation on shear capacity of RC beams with GFRP rebar and stirrups. *Steel Compos. Struct.* 21(6), 1265–1285 (2016). <https://doi.org/10.12989/scs.2016.21.6.1265>
- [59] Razaqpur, A. G., and Spadea, S.: Shear strength of FRP reinforced concrete members with stirrups. *J. Compos. Constr.* 19(1), 04014025 (2015). [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000483](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000483)
- [60] Marí, A., Cladera, A., Oller, E., and Bairán, J.: Shear design of FRP reinforced concrete beams without transverse reinforcement. *Compos. Part B Eng.* 57, 228–241 (2014). <https://doi.org/10.1016/j.compositesb.2013.10.005>
- [61] Feng, C., Lu, C., Wang, J., Qian, H., and Yan, Y.: Shear resistance and calculation method of GFRP reinforced concrete beams with CFRP mesh fabric as shear reinforcement. *Case Stud. Constr. Mater.* 21, e03904 (2024). <https://doi.org/10.1016/j.cscm.2024.e03904>
- [62] Shehata, I. A., and Regan, P. E.: Punching in reinforced concrete slabs. *J. Struct. Eng.* 115(7), 1726–1740 (2000). [https://doi.org/10.1061/\(ASCE\)0733-9445\(1989\)115:7\(1726\)](https://doi.org/10.1061/(ASCE)0733-9445(1989)115:7(1726))
- [63] Alam, M. S.: Influence of different parameters on shear strength of FRP reinforced concrete beams without web reinforcement. Ph.D. dissertation, Memorial University of Newfoundland, Canada (2010).

[64] Krall, M. D.: Tests on Concrete Beams with GFRP Flexural and Shear Reinforcements & Analysis Method for Indeterminate Strut-and-Tie Models with Brittle Reinforcements. M.A.Sc. thesis, University of Waterloo, Waterloo, Ontario, Canada (2014).